In Person Public Health Training

Interpreting Hepatitis B Serological Laboratory Results: Assessment

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Assessment

Part One: Knowledge Check (Pre and Post)

Below are ten questions over the training materials. This assessment is a knowledge check, and you may **NOT** use your handout during this portion of the training. Please circle one answer for each question. (The assessment is double sided. Please flip your paper to ensure you do not miss any questions.)

1. Which serologic laboratory results show that hepatitis B virus is currently present?
    a. Anti-HBs
    b. HBsAg
    c. Total IgM/IgG Anti-HBc
    d. Anti-HBe
2. A person who has received their full hepatitis B vaccine series and was never exposed to the virus would be expected to have what laboratory results?
    a. Anti-HBs + and Total IgM/IgG Anti-HBc +
    b. HBsAg + only
    c. Anti-HBs + only
    d. Total IgM/IgG Anti-HBc + only
3. What is the difference between screening and diagnostic testing?
    a. Screening uses antibody tests and diagnosis uses antigen tests
    b. Screening looks for infection in people without symptoms and diagnostic testing evaluates symptoms or a known exposure
    c. Only positive screening results are reportable
    d. There is no difference
4. IgM Anti-HBc + most often indicates
    a. A false positive HBsAg
    b. Lifelong chronic infection
    c. Recent infection within the last six months
    d. Immunity from vaccination

5. Per the CDC chronic hepatitis B is defined as an infection lasting at least:
    a. 2 weeks
    b. 6 weeks
    c. 6 months
    d. 1 year
6. Per the CDC 2024 case definition, which of the following is part of the clinical criteria for acute hepatitis B?
    a. AST >100 IU/L
    b. ALT >200 IU/L
    c. Nausea
    d. Anti-HBs +
7. The CDC recommended one-time adult screening triple panel includes:
    a. HBsAg, Anti-HBs, Total IgM/IgG Anti-HBc
    b. HBsAg, HBeAg, HBV DNA
    c. IgM anti-HBc, Anti-HBe
    d. HBsAg, AST, Bilirubin
8. A positive HBsAg a week after receiving a hepatitis B vaccine dose is most likely:
    a. Acute infection
    b. Chronic infection
    c. False positive
    d. Reactivating infection
9. A quantitative anti-HBs result of 5 mIU/mL after vaccination indicates a patient is:
    a. Immune
    b. Susceptible
    c. Infectious
    d. Indeterminate
10. Total IgM/IgG Anti-HBc + indicates a patient:
    a. Received their complete hepatitis B vaccine series
    b. Was infected with hepatitis B at some point in their life
    c. Is currently infectious
    d. Has received HBIG

Part Two: Group Case Review

Below are three patients with different patient history and laboratory results. In your groups review the patient history, review laboratory results, and do the following:

1. Interpret laboratory results
2. Assess relevant history
3. Classify cases
4. Identify next steps

This assessment is a group case review, and you **MAY** use your handout during this portion of the training. (The assessment is double sided. Please flip your paper to ensure you do not miss any questions.)

Patient One

Edward Jenner, 35-year-old man who injects drugs was seen in the emergency department with complaints of nausea and jaundice.

Laboratory results: HBsAg +, IgM Anti-HBc +, Anti-HBs -, ALT 999 IU/L

Patient Two

Florence Nightingale, 31-year-old woman at a routine sexual health testing, first vaccine dose seven days ago, asymptomatic

Laboratory results: HBsAg +, IgM Anti-HBc -, Anti-HBs -, ALT 10 IU/L

Patient Three

Johnny Blizzard, 50-year-old man on rituximab for rheumatoid arthritis, reported as chronic hepatitis B in 2016.

Laboratory results: HBsAg +, Total IgM/IgG Anti-HBc +, IgM Anti-HBc + (low), Anti-HBs -, ALT 500 IU/L

Answer Key

Part One:

B, C, B, C, C, B, A, C, B, B

Part Two:

Patient One

Acute infection, confirmed with laboratory evidence, history of injection drug use, and jaundice. Interview provider and patient, identify any shared drug use and sexual partners, test and vaccinate possible susceptible contacts, link to provider for follow-up, retest HBsAg at six months

Patient Two

False positive patient, tested after vaccination, no exposure, no symptoms, screening labs positive, not a case, Anti-HBs negative due to vaccination first in series, body has not produced antibodies yet.

Patient Three

Rheumatic medication carries likelihood of reactivating chronic hepatitis B patients, not an acute case, chronic confirmed patient from 2016 already reported to CDC per reporting guidelines, notify patient provider about reactivation risk given immunosuppressant effect of rheumatic medication.

Part Two Rubric:

	Full (2 points)	Partial (1 point)	None (0 points)
Interpretation	Correct w/labs	Partial w/labs	Incorrect
History	Relevant history	Partial	Incorrect
Classification	Correct status and type	Correct status/wrong type	Incorrect
Next Steps	Specific/Appropriate	Appropriate/Vague	Incorrect